Refactoring to Eclipse Collections

Making Your Java Streams Leaner, Meaner, and Cleaner

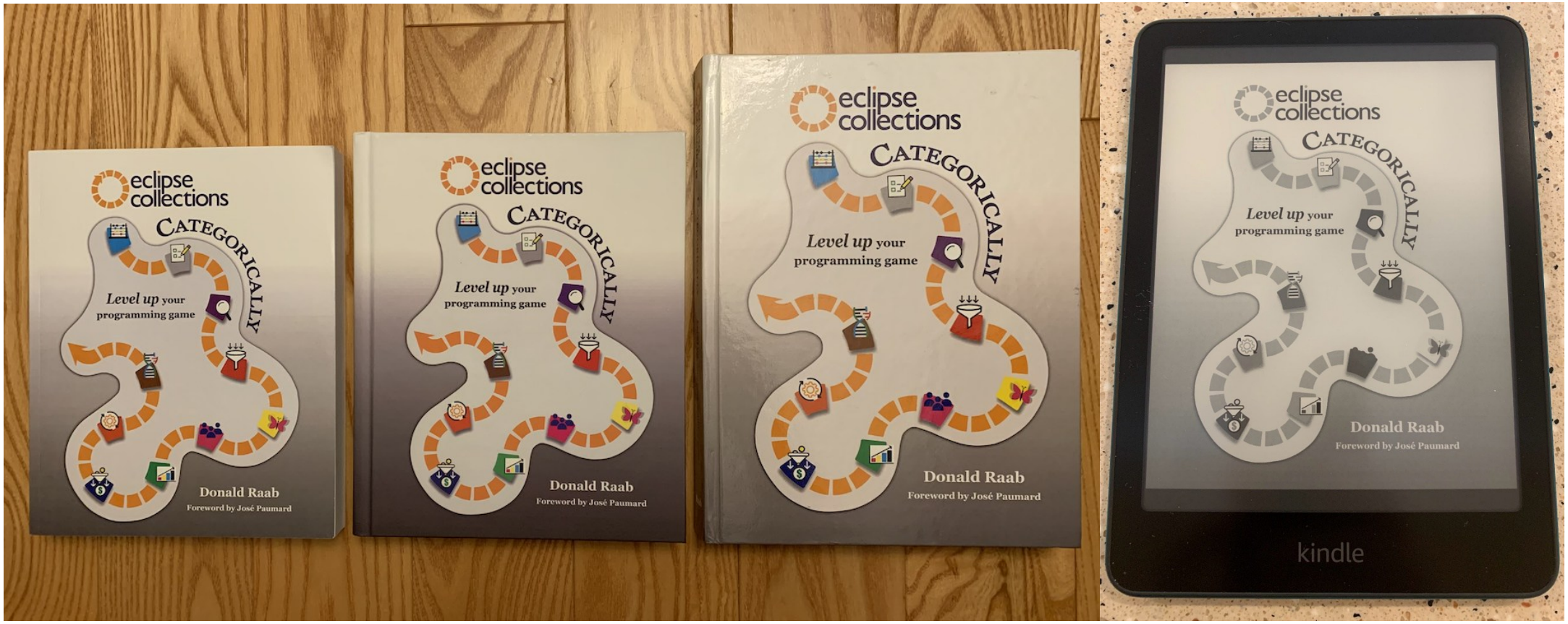
Introduction

- What is Eclipse Collections?
  - Feature rich, memory efficient Java Collections framework
- History
  - Eclipse Collections started off as an internal collections framework named Caramel at Goldman Sachs in 2004

  - In 2012, it was open sourced to GitHub as a project called GS Collections
  - GS Collections was migrated to the Eclipse Foundation, re-branded as Eclipse Collections in 2015
  - Eclipse Collections 12.0 (Java 11+) & 13.0 (Java 17+) released in June 2025
- Learn Eclipse Collections with Kata
- Eclipse Collections is open for contributions!

Book: Eclipse Collections Categorically

- Paperback, Hardcover(x2), and eBook Versions Available
- Amazon / Barnes & Noble



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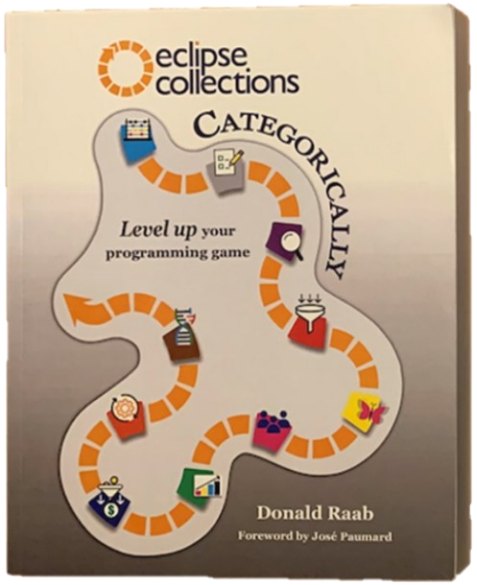
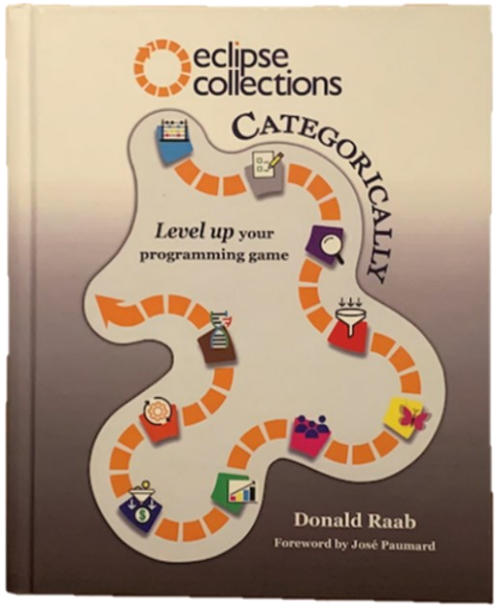
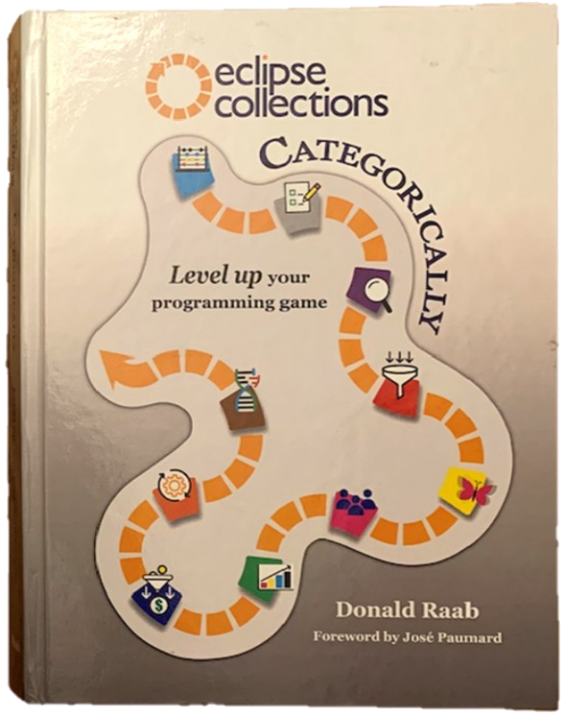
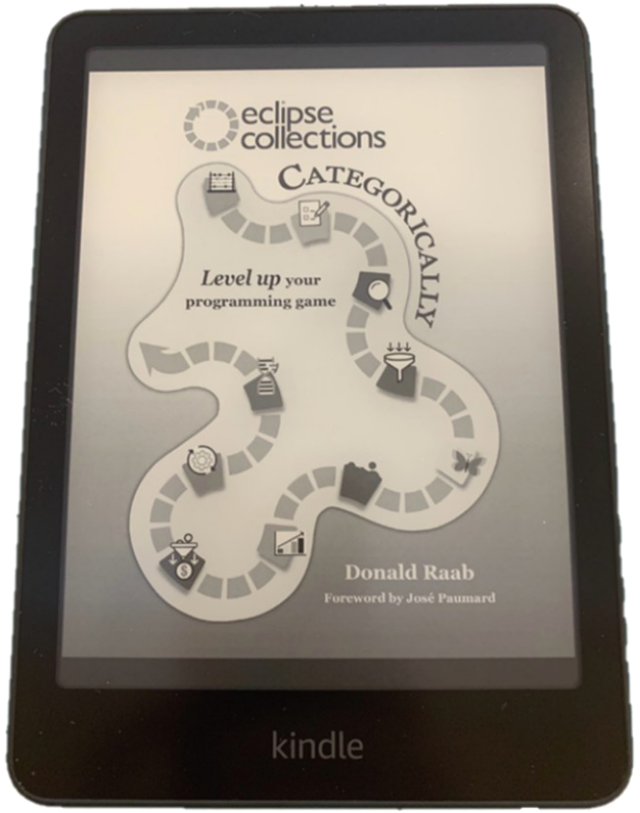




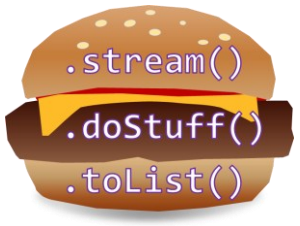




Simulating Method Categories in Java

- Use IntelliJ Custom Code Folding Regions

Why Refactor to EC?



Lazy or Eager

TestUtils

Testability/Debuggability

No “Bun” methods

Drop-in replacement for JDK types

Concise Code

Benefits

It’s Easy!

Intuitive humane APIs

Rich[er] APIs

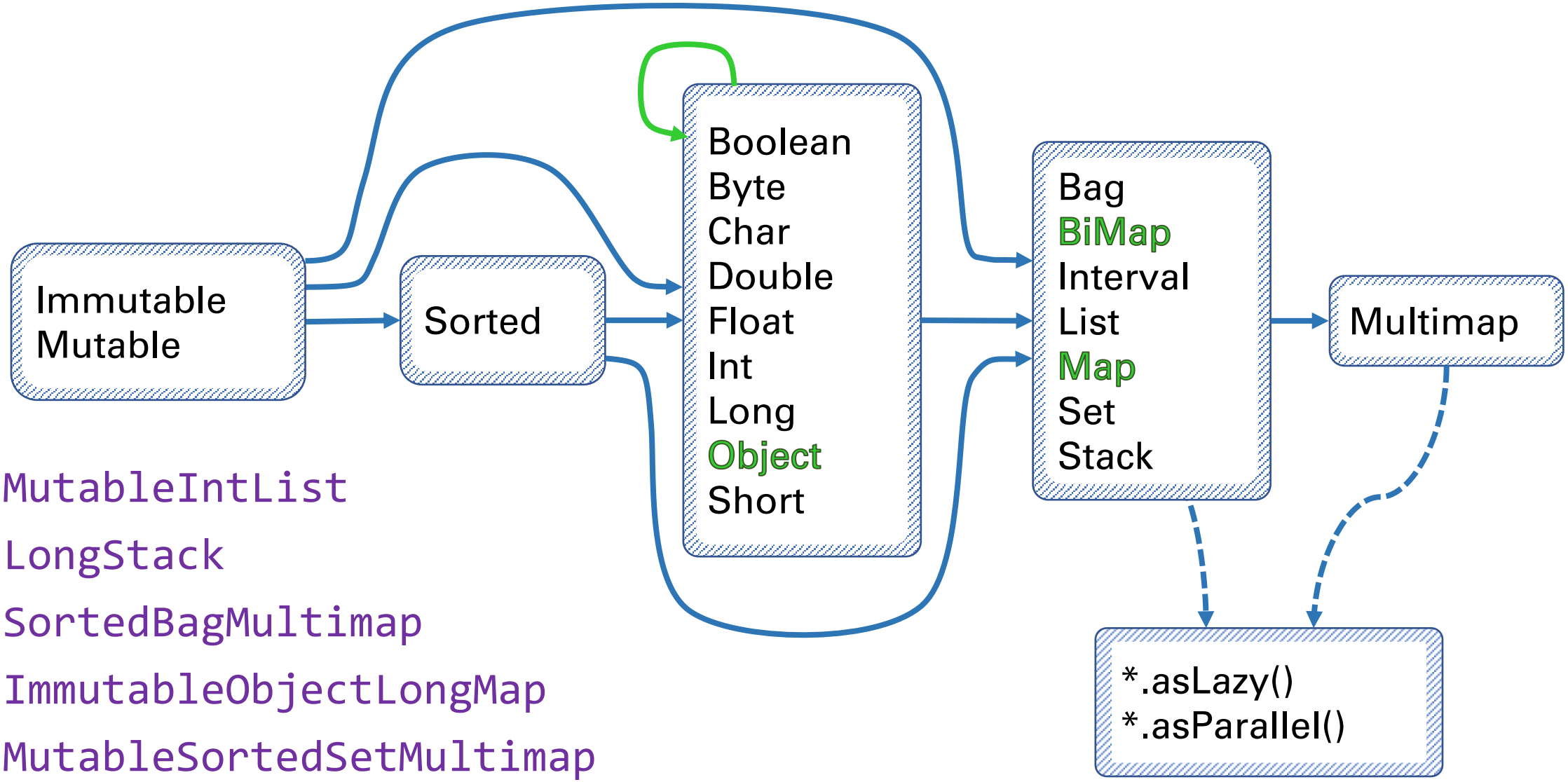
Memory Savings

Better Performance [*citation needed*]

Comprehensive Support for Primitive Types

Efficient Maps and Sets

Any Types You Need



MutableIntList

LongStack

SortedBagMultimap

ImmutableObjectLongMap

MutableSortedSetMultimap

Instantiate Them Using Factories

Primitive Type	Container Type	Mutability	Multimap	Initialized	Lazy
Boolean	Bags	.mutable	.bag	.empty()|.of()|.with()	.asLazy()
Byte	BiMaps	.immutable	.list	.of(one)|.with(one)	
Char	Lists	.fixedSize	.set	...	
Double	Maps		.sortedSet	.of(one,…,ten)	
Float	Multimaps			.with(one,…,ten)	
Int	Sets			.of(... elements)	
Long	SortedBags			.with(... elements)	
Object	SortedMaps			.ofAll(Iterable)	
Short	SortedSets			.withAll(Iterable)	
	Stacks				

```
ImmutableLongStack
LongStacks.immutable.with(1, 2, 3).asLazy()
LazyLongIterable
```

Methods by Category - Highlights

transform

collect[With]
collect[Boolean,Byte,Char,Double,Float,Int,Long,Short]
collectIf
collectKeysAndValues
collectValues
collectWithIndex
collectWithOccurrences
flatCollect

group

groupBy
groupByEach
groupByUniqueKey
sumBy[Double,Float,Int,Long]
sumOf[Double,Float,Int,Long]
aggregateBy
aggregateInPlaceBy

convert

toArray
toBag
toImmutable
toList
toMap
toReversed
toSet
toSortedBag[By]
toSortedList[By]
toSortedMap
toSortedSet
toSortedSet[By]
toStack
toString

filter

select[With]
selectByOccurrences
selectInstancesOf
reject[With]
partition[With]
partitionWhile

wrap

asLazy
asParallel
asReversed
asSynchronized
asUnmodifiable

find

detect[With]
detect[With]IfNone
detect[With]Optional
max[By]
min[By]

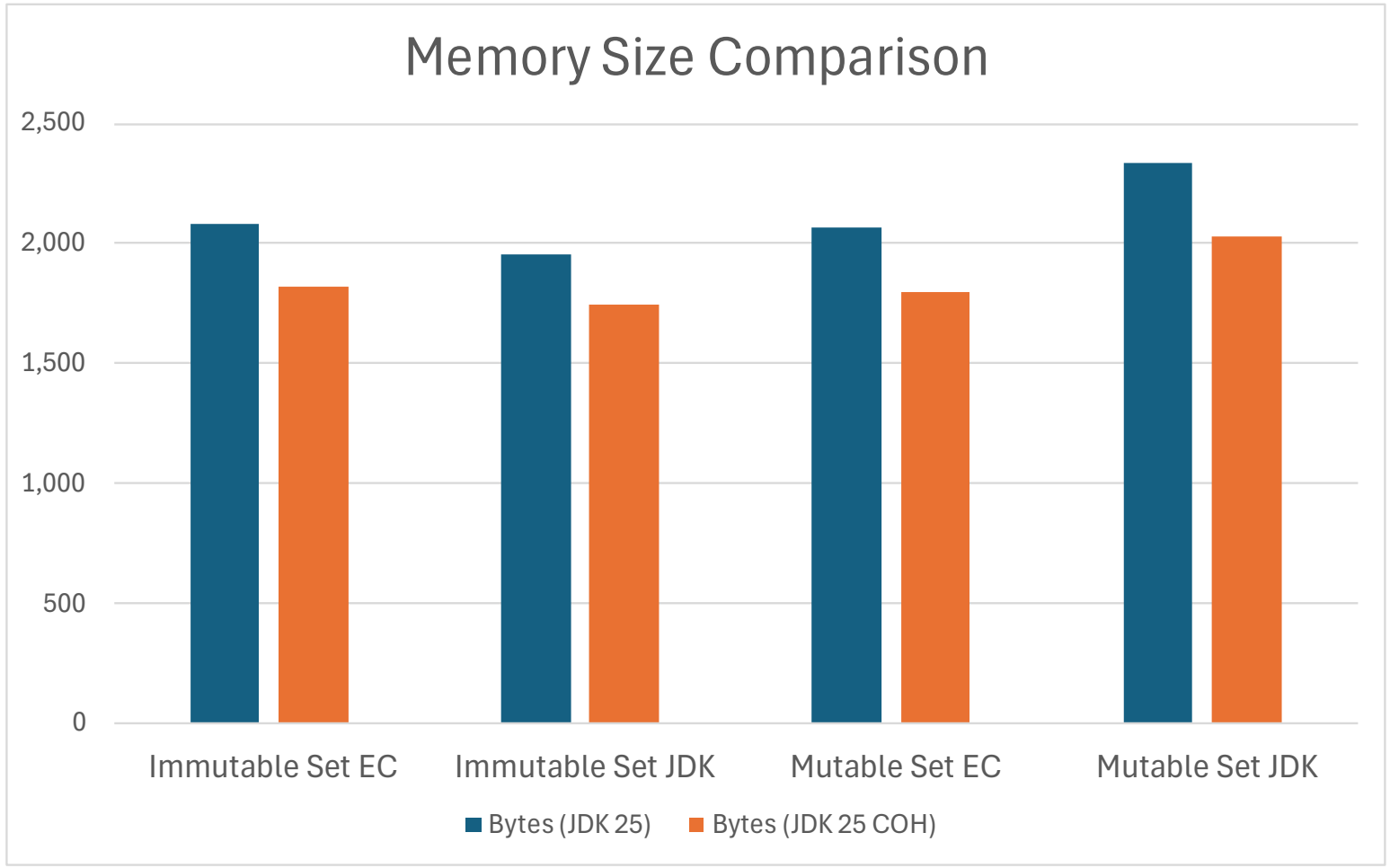
test

allSatisfy[With]
anySatisfy[With]
noneSatisfy[With]
notEmpty
isEmpty

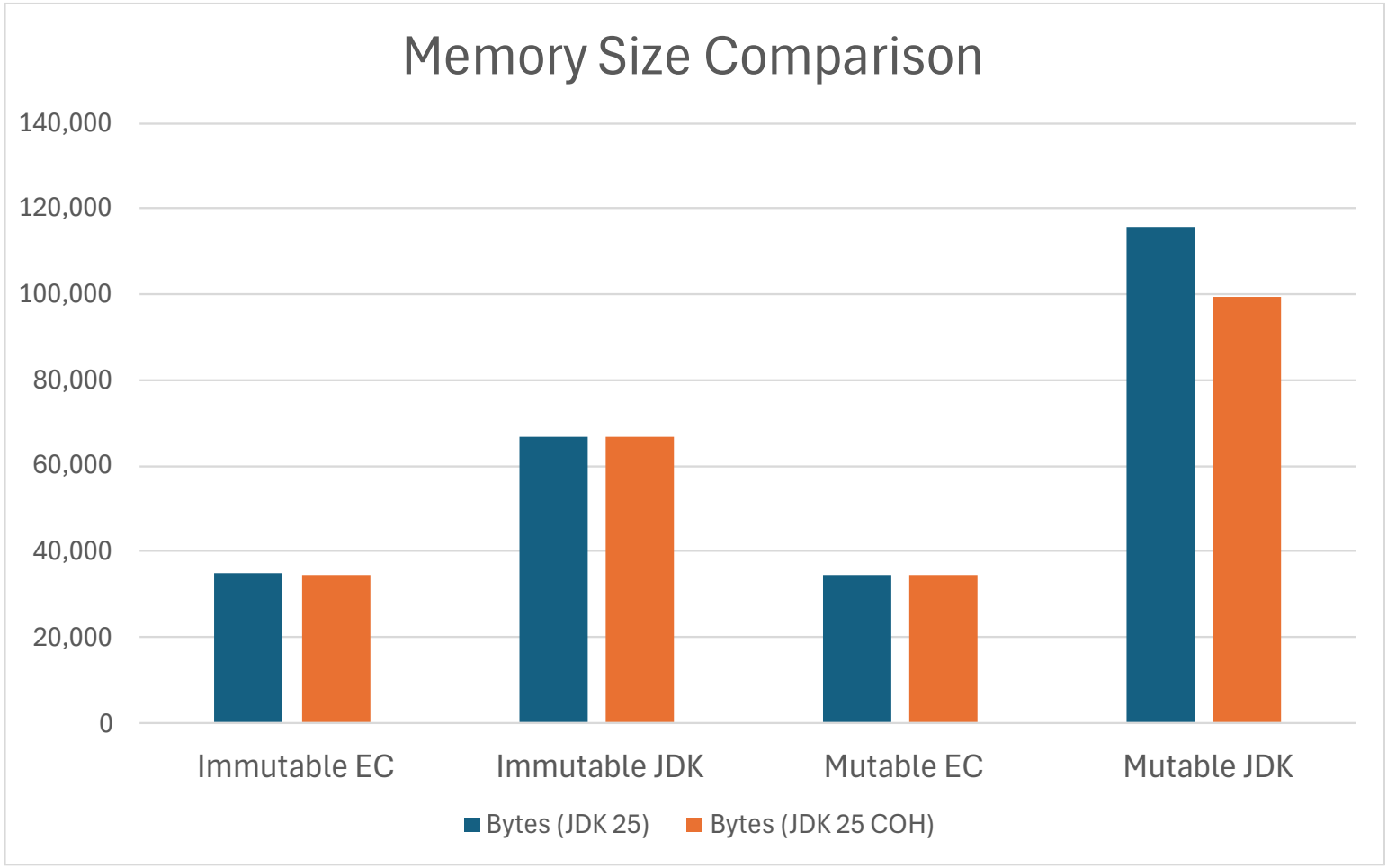
Methods – Lots More…

Let’s Do It!

Memory: Generation Set



Memory: Generations By Year



Memory Usage (Overhead in KB, Count)